

```

SELECT  AVG(subquery.tot)
FROM    ( SELECT  A.custid, SUM(A.balance) AS tot
        FROM      Account A
        GROUP BY  A.custid ) AS subquery ;

```

CAST( 102.4675 AS NUMERIC(5,2) ) gives 102.47

Useful also to produce values in a specific format

### Axiomatization (records)

**Reflexivity:**  $R[X] \subseteq R[X]$

**Transitivity:** If  $R[X] \subseteq S[Y]$  and  $S[Y] \subseteq T[Z]$ , then  $R[X] \subseteq T[Z]$

**Projection:** If  $R[X, Y] \subseteq S[W, Z]$  with  $|X| = |W|$ , then  $R[X] \subseteq S[W]$

**Permutation:** If  $R[A_1, \dots, A_n] \subseteq S[B_1, \dots, B_n]$ , then  $R[A_{i_1}, \dots, A_{i_n}] \subseteq S[B_{i_1}, \dots, B_{i_n}]$ , where  $i_1, \dots, i_n$  is a permutation of  $1, \dots, n$

### Essential axioms (set of attributes)

**Reflexivity:** If  $Y \subseteq X$ , then  $X \rightarrow Y$

**Augmentation:** If  $X \rightarrow Y$ , then  $XZ \rightarrow YZ$  for any  $Z$

**Transitivity:** If  $X \rightarrow Y$  and  $Y \rightarrow Z$ , then  $X \rightarrow Z$

### Other axioms

**Union:** If  $X \rightarrow Y$  and  $X \rightarrow Z$ , then  $X \rightarrow YZ$

**Decomposition:** If  $X \rightarrow YZ$ , then  $X \rightarrow Y$  and  $X \rightarrow Z$

Minimal Cover = 1) un seul Attr à droite

2) On peut pas enlever une FD et tjrs avoir une cover

3) Chaque FD a le nb minimal d'attributs à gauche

1. **First Law:**  $\neg(\forall x P(x)) \equiv \exists x \neg P(x)$

2. **Second Law:**  $\neg(\exists x P(x)) \equiv \forall x \neg P(x)$

### Projection

$\pi_\alpha(E)$  is translated to  $\exists X \varphi$

where

- $\varphi$  is the translation of  $E$
- $X = \text{free}(\varphi) - \eta(\alpha)$   
(attributes that are **not** projected become quantified)

Two operations (from different transactions) are conflicting if

- they refer to the same data item, and
- at least one of them is a write

In a schedule, two consecutive non-conflicting operations (from different transactions) can be swapped

A schedule is conflict serializable if it can be transformed into a serial schedule by a sequence of swap operations

### Selection

$\sigma_\theta(E)$  is translated to  $\varphi \wedge \eta(\theta)$

where

- $\varphi$  is the translation of  $E$
- $\eta(\theta)$  is obtained from  $\theta$  by replacing each attribute  $A$  by  $\eta(A)$

A relation with FDs  $\Sigma$  is in **BCNF** if for every  $X \rightarrow Y$  in  $\Sigma$ :

- $Y \subseteq X$  (the FD is trivial), or
- $X$  is a key

to find an equivalent serial schedule we still need to construct the precedence graph. For lock-based schedules, this is constructed by examining each lock, finding the corresponding unlock operation by the same transaction, say  $T_i$ , and looking for a conflicting lock subsequently obtained by a different transaction  $T_j$ : if that is the case, there is an edge from  $T_i$  to  $T_j$ .

For  $R$  over attributes  $A_1, \dots, A_n$

$\text{adom}(R)$  is given by  $\rho_{A_1 \rightarrow A}(\pi_{A_1}(R)) \cup \dots \cup \rho_{A_n \rightarrow A}(\pi_{A_n}(R))$

$\text{adom}(D) = \bigcup_{R \in D} \text{adom}(R)$

```

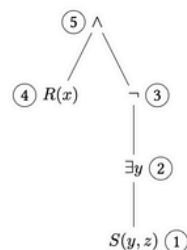
SELECT DISTINCT C.id, C.name
FROM   Customer C JOIN Account A ON C.id = A.custid
WHERE  A.balance >= ALL ( SELECT balance
                        FROM   Account );

```

**Problem 2.** Given a relation  $R$  over attribute  $A$ , and a relation  $S$  over attributes  $A, B$  (in this order), translate the following relation calculus query to relational algebra:

$\{x, z \mid R(x) \wedge \neg \exists y S(y, z)\}$

**Solution.** The syntax tree of the (body of the) given relational calculus query is:



Under the mapping  $\eta = \{x \mapsto A, y \mapsto B, z \mapsto C\}$ , we get the following translations of each subexpression (corresponding to the subtrees above labelled with circled numbers):

(1)  $\mapsto \rho_{B \rightarrow C, A \rightarrow B}(S)$

(2)  $\mapsto \pi_C((1))$

(3)  $\mapsto \text{Adom}_C - (2)$

(4)  $\mapsto R$

(5)  $\mapsto ((4) \times \text{Adom}_C) \cap ((3) \times \text{Adom}_A)$

Putting it all together, we obtain the following:

$(R \times \text{Adom}_C) \cap (\text{Adom}_A \times (\text{Adom}_C - \pi_C(\rho_{B \rightarrow C, A \rightarrow B}(S)))$

Suppose  $T_i$  requests a lock and  $T_j$  holds a conflicting lock

Two policies to prevent deadlocks:

**Wait-die:**  $T_i$  waits if it has higher priority, otherwise **aborted**

**Wound-wait:**  $T_i$  **aborted** if  $T_i$  has higher priority, otherwise  $T_i$  waits

### 2PL Strict:

- The schedule is always recoverable
- All aborted transactions can be rolled back without cascading aborts
- The schedule consisting of the committed transactions is conflict serializable

Given a set  $\Sigma$  of FDs on attributes  $U$ , how do we compute all candidate keys?

1.  $ck := \emptyset$

2.  $G :=$  DAG of the powerset  $2^U$  of  $U$

- Nodes are elements of  $2^U$  (sets of attributes)
- There is an edge from  $X$  to  $Y$  if  $Y \subseteq X$  and  $X - Y = \{A\}$

3. Repeat until  $G$  is empty:

Find a node  $X$  without children

if  $C_\Sigma(X) = U$ :

$ck := ck \cup \{X\}$

Delete  $X$  and all its ancestors from  $G$

else:

Delete  $X$  from  $G$

**Customer** : CustID, Name

**Account** : Number, CustID

Translate  $\exists x_4 \text{ Customer}(x_1, x_2) \wedge \text{Account}(x_3, x_4) \wedge x_1 = x_4$

Environment  $\eta = \{x_1 \mapsto A, x_2 \mapsto B, x_3 \mapsto C, x_4 \mapsto D\}$

$\pi_{A,B,C}((E_1 \times \text{Adom}_C \times \text{Adom}_D) \cap (\text{Adom}_A \times \text{Adom}_B \times E_2) \cap (\sigma_{A=D}(\text{Adom}_A \times \text{Adom}_D) \times \text{Adom}_B \times \text{Adom}_C))$

where

$E_1 = \rho_{\text{CustID} \rightarrow A, \text{Name} \rightarrow B}(\text{Customer})$

$E_2 = \rho_{\text{Number} \rightarrow C, \text{CustID} \rightarrow D}(\text{Account})$

## 3NF synthesis algorithm

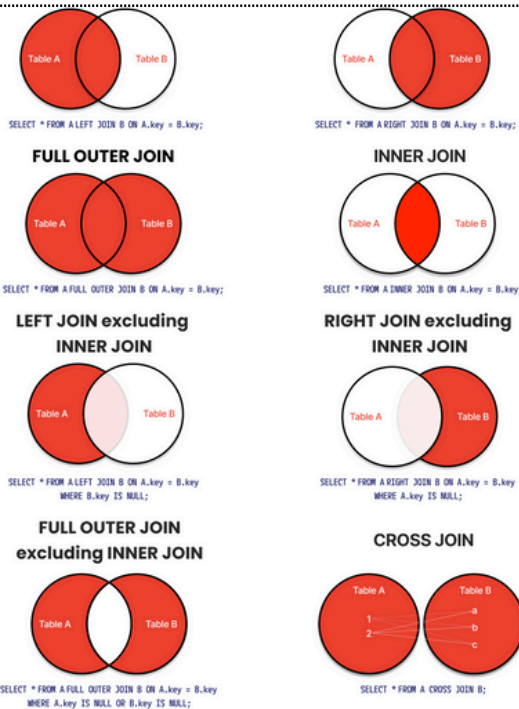
$R(x_1, \dots, x_n)$  is translated to  $\rho_{A_1 \rightarrow \eta(x_1), \dots, A_n \rightarrow \eta(x_n)}(R)$

**Input** : A set of attributes  $U$  and a set of FDs  $\Sigma$

**Output** : A database schema  $S$

1.  $S := \emptyset$
2. Find a minimal cover  $\Gamma$  of  $\Sigma$
3. Replace all FDs  $X \rightarrow A_1, \dots, X \rightarrow A_n$  in  $\Gamma$  by  $X \rightarrow A_1 \dots A_n$  (note that the FDs have the same l.h.s.)
4. For each FD  $(X \rightarrow Y) \in \Gamma$ , add  $(XY, X \rightarrow Y)$  to  $S$
5. If no  $(U_i, \Sigma_i)$  in  $S$  is such that  $U_i$  is key for  $(U, \Sigma)$ , find a key  $K$  for  $(U, \Sigma)$  and add  $(K, \emptyset)$  to  $S$
6. If  $S$  contains  $(U_i, \Sigma_i)$  and  $(U_j, \Sigma_j)$  with  $U_i \subseteq U_j$ , replace them by  $(U_j, \Sigma_i \cup \Sigma_j)$
7. Output  $S$

```
CREATE TRIGGER invoice_order
AFTER UPDATE OF final ON orders
REFERENCING OLD ROW AS oldrow
NEW ROW AS newrow
FOR EACH ROW
WHEN oldrow.final = FALSE AND newrow.final = TRUE
BEGIN
    INSERT INTO invoices(ordid, amount, issued, due)
    SELECT O.ordid, SUM(D.qty * P.price),
           O.odate, O.odate+7d
    FROM   orders O, details D, prices P
    WHERE  O.ordid = newrow.ordid
    AND    O.ordid = D.ordid
    AND    D.ordid = P.ordid
    AND    D.pcode = P.pcode
END ;
```



## Negation

$\neg \varphi$  is translated into  $(\bigtimes_{x \in \text{free}(\varphi)} \text{Adom}_{\eta(x)}) - E$

where  $E$  is the translation of  $\varphi$

## Finding minimal covers

1. Put the FDs in standard form: only one attribute on RHS  
Use Armstrong's decomposition axiom

$X \rightarrow A_1 \dots A_n$  is split into  $n$  FDs:  $X \rightarrow A_1, \dots, X \rightarrow A_n$

2. Minimize the LHS of each FD  
Check whether attributes in the LHS can be removed

For  $(X \rightarrow A) \in \Sigma$  and  $X' \subset X$  check whether  $A \in C_{\Sigma}(X')$   
If yes, replace  $X \rightarrow A$  by  $X' \rightarrow A$  and repeat

3. Delete redundant FDs  
 $(X \rightarrow A) \in \Sigma$ , check whether  $\Sigma - \{X \rightarrow A\} \models X \rightarrow A$

## Existential quantification

$\exists x \varphi$  is translated to  $\pi_{\eta(x - \{x\})}(E)$

where

- $E$  is the translation of  $\varphi$
- $X = \text{free}(\varphi)$

**Disjunction:**  $\varphi_1 \vee \varphi_2$  is translated to

$$E_1 \times \left( \bigtimes_{x \in X_2 - X_1} \text{Adom}_{\eta(x)} \right) \cup E_2 \times \left( \bigtimes_{x \in X_1 - X_2} \text{Adom}_{\eta(x)} \right)$$

where, for  $i \in \{1, 2\}$ ,

**Problem 3.** Given two relations  $R$  and  $S$ , each over attributes  $A, B$  (in this order), express the following relational calculus query in relational algebra:

$$\{x \mid \neg(\forall y R(x, y) \rightarrow S(x, y)) \wedge \neg(\exists z S(x, z) \wedge R(z, x))\}$$

$$\pi_A(R \cap (\text{Adom}_A \times \text{Adom}_B - S)) \cap (\text{Adom}_A - \pi_A(\rho_{B \rightarrow C}(S) \cap \rho_{A \rightarrow C, B \rightarrow A}(R)))$$

## From algebra to calculus

**Renaming**  $\rho_{\text{old} \rightarrow \text{new}}(E)$

1. Translate  $E$  to  $\varphi$
2. If there is no mapping for **new** in  $\eta$ , add  $\{\text{new} \mapsto x_{\text{new}}\}$
3. Replace every occurrence of  $\eta(\text{new})$  in  $\varphi$  with a **fresh** variable
4. Replace every (free) occurrence of  $\eta(\text{old})$  in  $\varphi$  by  $\eta(\text{new})$

**Theta-join**

$$R \bowtie_{\theta} S \equiv \sigma_{\theta}(R \times S)$$

**Equijoin**

$\bowtie_{\theta}$  where  $\theta$  is a **conjunction of equalities**

**Semijoin**

$$R \ltimes_{\theta} S \equiv \pi_X(R \bowtie_{\theta} S)$$

where  $X$  is the set of attributes of  $R$

**Antijoin**

$$R \bar{\bowtie}_{\theta} S \equiv R - (R \ltimes_{\theta} S)$$

ACCOUNT  $\bar{\bowtie}_{\text{Balance} < \text{Balance}'}$  ACCOUNT'

Gives all the accounts whose balance is less than that of no other account